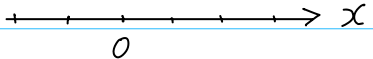


Brief review: motion in one dimension 

position x velocity $v = \frac{dx}{dt}$ acceleration $a = \frac{dv}{dt}$

1D constant a motion ($a = a_0$) kinematic equations:

$$v_f = a_0 t + v_i \quad (1)$$

$$x_f = \frac{1}{2} a_0 t^2 + v_i t + x_i \quad (2)$$

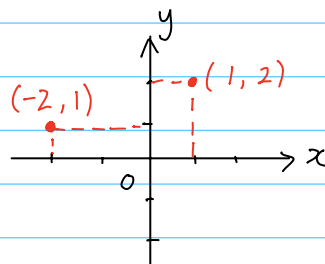
$$x_f = \frac{1}{2} (v_f + v_i) t + x_i \quad (3)$$

$$v_{avg} = \frac{1}{2} (v_f + v_i) \quad (4)$$

$$v_f^2 = v_i^2 + 2 a_0 (x_f - x_i) \quad (5)$$

3.1 2D Coordinate systems

①



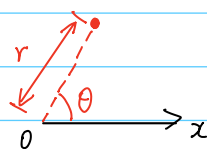
like table top, blackboard surface, etc.

position of a particle • written as (x, y)

you can find x by projecting the particle on the x -axis
you can find y by projecting the particle on the y -axis

"Cartesian coordinate system" or "rectangular coordinate system".

② Alternatively we can represent the position as (r, θ)

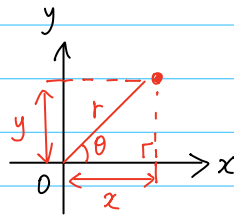


r is the distance from origin ($r \geq 0$)

θ is the angle between the origin-particle line and the polar axis (x -axis in this case)

"polar coordinate system"

Conversion between Cartesian and polar coordinate systems:



Use trigonometric functions in this right triangle.

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan\left(\frac{y}{x}\right) \end{cases}$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

3.2

scalar e.g. mass, distance, speed, temperature, length, volume, ...

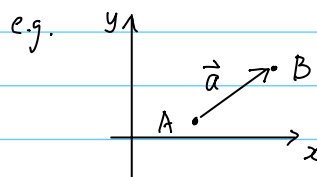
only magnitude (positive or negative), no direction

symbol: plain letter e.g. $m = 5 \text{ kg}$

vector e.g. displacement, velocity, acceleration, force, electric field ...

has both magnitude and direction

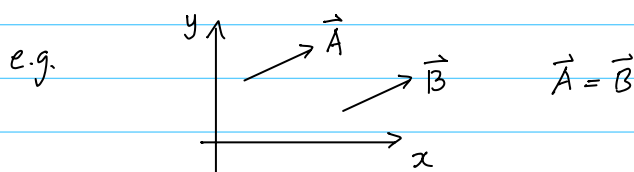
symbol: letter with arrow hat or bold font (when typing)



vector $\vec{a}$ $\begin{cases} \text{magnitude: length of arrow, } 5 \text{ m} \\ \text{direction: arrow direction, from A to B} \end{cases}$

3.3 Vector operations.

① Equal $\vec{A} = \vec{B}$ when both the magnitude and direction are the same.



"translational invariant"
(In physics, translation means moving without rotation)

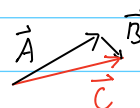
② Addition $\vec{A}$ $\vec{B}$

Define $\vec{A} + \vec{B} =$

step 1. move $\vec{B}$ so that the starting point of $\vec{B}$ is at the end point of $\vec{A}$

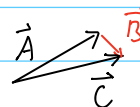


step 2. draw an arrow from the starting point of $\vec{A}$ to the end point of $\vec{B}$



This is the result, $\vec{C} = \vec{A} + \vec{B}$ ("head to tail method")

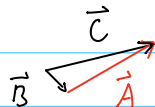
③ Subtraction From above, $\vec{B} = \vec{C} - \vec{A}$



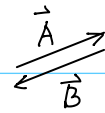
step 1. join the starting points of the two vectors ($\vec{A}$ and $\vec{C}$),

step 2. the result ($\vec{B}$) is pointing
from the end point of the second vector ($\vec{A}$)
to the end point of the first vector ($\vec{C}$)

We can see that $\vec{A} = \vec{C} - \vec{B}$ is the same,

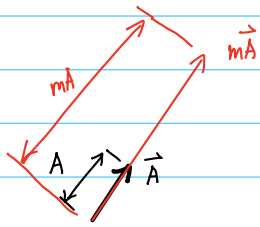


(3') Negative from (3), let $\vec{C} = 0$, $\vec{B} = -\vec{A}$



same magnitude reverse direction

(4) scalar multiplication scalar m times vector $\vec{A}$



result $\vec{R} = m\vec{A}$ is a vector with magnitude $|mA|$ and direction same as $\vec{A}$

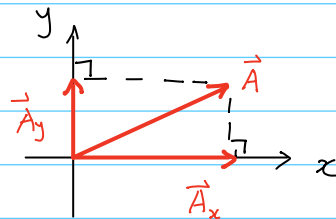
if $m < 0$, $\vec{R}$ and $\vec{A}$ have opposite direction

3.4

Any vector $\vec{A}$ in two dimension can be decomposed into two components

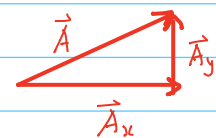
$\vec{A}_x$ and $\vec{A}_y$ in a Cartesian coordinate system.

e.g.



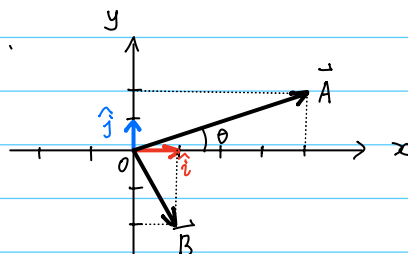
projection on each axis

$$\vec{A} = \vec{A}_x + \vec{A}_y$$



unit vector is a vector with unit length (i.e. magnitude = 1).
is useful to define the direction.
symbol = use a hat ^ instead of arrow

e.g.



$\rightarrow \hat{i}$ and $\uparrow \hat{j}$ are unit vectors.

$$\vec{A} = 4\hat{i} + 2\hat{j}$$

$$\vec{B} = \hat{i} - 2\hat{j}$$

Using unit vectors, we can write $\vec{A} = \vec{A}_x + \vec{A}_y$ as $\vec{A} = A_x \hat{i} + A_y \hat{j}$

$$A_x = |\vec{A}| \cos \theta \quad (\theta \text{ is the angle between } \vec{A} \text{ and } \hat{i})$$

$$A_y = |\vec{A}| \sin \theta$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

Note that x and y components are orthogonal to each other, they are independent.

When we do additions, we can add each components independently.

e.g. $\vec{C} = \vec{A} + \vec{B}$

$$C_x \hat{i} + C_y \hat{j} = (A_x \hat{i} + A_y \hat{j}) + (B_x \hat{i} + B_y \hat{j})$$

$$C_x \hat{i} + C_y \hat{j} = (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j}$$

Compare each component, we have

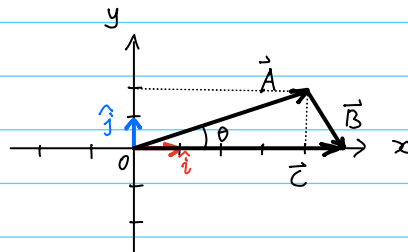
$$C_x = A_x + B_x$$

$$C_y = A_y + B_y$$

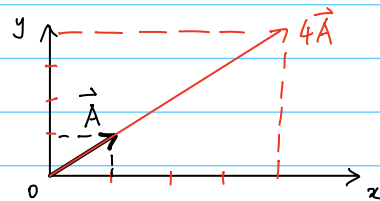
e.g. For the $\vec{A}$ and $\vec{B}$ above, $\vec{A} = 4\hat{i} + 2\hat{j}$

$$\vec{B} = \hat{i} - 2\hat{j}$$

$$\text{the } \vec{C} = \vec{A} + \vec{B} = (4+1)\hat{i} + (2-2)\hat{j} = 5\hat{i}$$

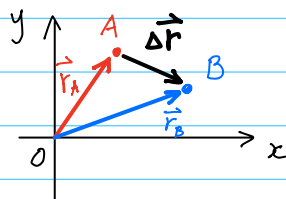


For scalar multiplication, $m\vec{A} = mA_x \hat{i} + mA_y \hat{j}$



4.1

Describe the position in two dimension using a position vector $\vec{r}$, pointing from origin to that position.



e.g. point A position $\vec{r}_A$

A particle is at point A when $t = t_A$, after a while, it moves to point B when $t = t_B$.

During this time, the displacement of the particle is :

$$\Delta \vec{r} = \vec{r}_B - \vec{r}_A$$

The average velocity in this time duration is :

$$\vec{v}_{\text{avg}} \equiv \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}_B - \vec{r}_A}{t_B - t_A}$$

The instantaneous velocity at a specific time point is when Δt approached zero at this moment.

$$\vec{v} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

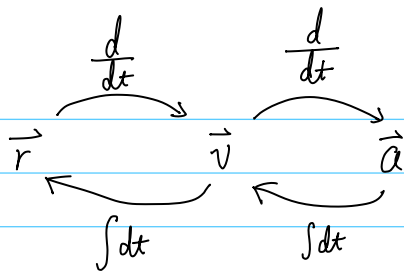
(instantaneous speed is the magnitude of $\vec{v}$, speed = $|\vec{v}|$)

Average acceleration during this time Δt is :

$$\vec{a}_{\text{avg}} \equiv \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_B - \vec{v}_A}{t_B - t_A}$$

Instantaneous acceleration at a specific moment is :

$$\vec{a} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$



same as in one dimension,
except that position, velocity, acceleration
are vectors in 2D

Note : To take the derivative of a vector $\vec{A}(t)$ with respect of time,
since $\vec{A}$ can be decomposed into two independent components
 $\vec{A}(t) = \vec{A}_x(t) + \vec{A}_y(t)$,
the derivative can be taken on each component.

$$\frac{d\vec{A}(t)}{dt} = \frac{dA_x(t)}{dt} \hat{i} + \frac{dA_y(t)}{dt} \hat{j}$$

Integral is similar:

$$\int \vec{A}(t) dt = \left(\int A_x(t) dt \right) \cdot \hat{i} + \left(\int A_y(t) dt \right) \cdot \hat{j}$$

★ Since any two-dimensional vectors can be decomposed into two orthogonal components, meaning that these two component are independent, any motion in two dimension can be decomposed into two one-dimensional motions in two orthogonal directions, which can be independently analysed.

4.2

motion in two dimension with constant acceleration

$$\vec{a} = \text{constant} \quad (\text{both magnitude and direction are constant})$$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} \quad , \quad a_x \text{ and } a_y \text{ are constants.}$$

On the x-axis, constant acceleration motion in one dimension

$$\begin{cases} a_x = \text{constant} \\ v_x = a_x t + v_{ix} \\ r_x = \frac{1}{2} a_x t^2 + v_{ix} t + r_{ix} \end{cases}$$

On the y-axis, constant acceleration motion in one dimension

$$\begin{cases} a_y = \text{constant} \\ v_y = a_y t + v_{iy} \\ r_y = \frac{1}{2} a_y t^2 + v_{iy} t + r_{iy} \end{cases}$$

$$\begin{aligned} \therefore \vec{v} &= (a_x t + v_{ix}) \hat{i} + (a_y t + v_{iy}) \hat{j} \\ &= \underline{(a_x \hat{i} + a_y \hat{j}) \cdot t} + (v_{ix} \hat{i} + v_{iy} \hat{j}) \quad \text{rearrange according to } t \\ &= \vec{a} t + \vec{v}_i \end{aligned}$$

$$\begin{aligned} \vec{r} &= \left(\frac{1}{2} a_x t^2 + v_{ix} t + r_{ix} \right) \cdot \hat{i} + \left(\frac{1}{2} a_y t^2 + v_{iy} t + r_{iy} \right) \cdot \hat{j} \\ &= \frac{1}{2} \underline{(a_x \hat{i} + a_y \hat{j}) \cdot t^2} + \underline{(v_{ix} \hat{i} + v_{iy} \hat{j}) \cdot t} + (r_{ix} \hat{i} + r_{iy} \hat{j}) \\ &= \frac{1}{2} \vec{a} t^2 + \vec{v}_i t + \vec{r}_i \end{aligned}$$

just like the one-dimensional form! but replaced with vectors.

4.3 Projectile motion

An object with an initial velocity, and moves under the sole effect of gravity.

$$\vec{a} = -g \hat{j} \quad (\hat{j} \text{ points upwards, } \vec{a} \text{ points downwards})$$

$$g \approx 9.8 \text{ m/s}^2$$

On x direction:

$$v_{xi} = v_i \cos \theta_i$$

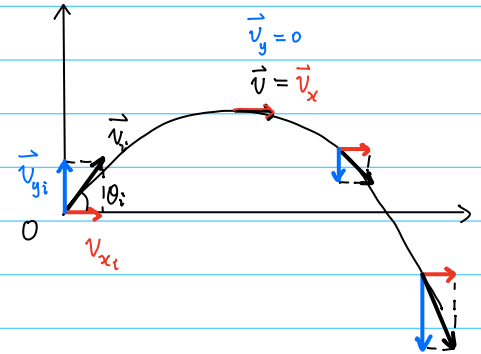
$$a_x = 0$$

constant velocity motion

$$v_x(t) = v_{xi}$$

$$x(t) = v_{xi} t + x_i$$

$$\rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \vec{v}_x$$



On y direction:

$$v_{yi} = v_i \sin \theta_i$$

$$a_y = -g$$

constant acceleration motion

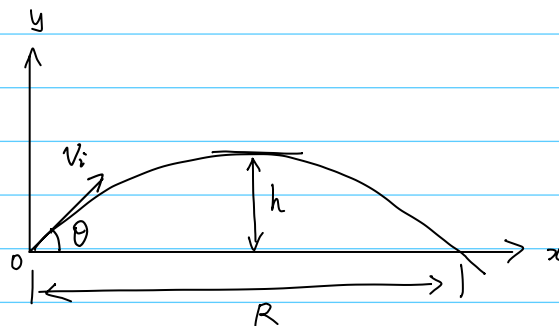
$$v_y(t) = -gt + v_{yi}$$

$$y(t) = \frac{1}{2} g t^2 + v_{yi} t + y_i$$

$$\begin{array}{c} \uparrow \downarrow \\ \uparrow \downarrow \\ \uparrow \downarrow \\ \downarrow \vec{v}_y \end{array}$$

e.g.1. A ball is kicked from ground with initial speed v_i pointing θ above ground.
Find (a) maximum height and (b) horizontal range. (use gravitational acceleration g)

Set up coordinate.



(a) In the y direction, maximum height is where $v_y = 0$ (and then flip sign).

★ identify which moments are "initial" and "final".

The question becomes $\Rightarrow$

Consider a one-dimensional constant acceleration problem in the y direction, given initial velocity $= v_{iy}$, final velocity $= 0$, $a = -g$, find displacement.

Use equation (5) from Lecture 2. $2a(x_f - x_i) = v_f^2 - v_i^2$ now in y direction.

$$\Rightarrow -2gh = 0^2 - v_{iy}^2$$

$$h = \frac{v_{iy}^2}{2g}, \text{ plug in } v_{iy} = v_i \sin \theta,$$

$$\boxed{h = \frac{v_i^2 \sin^2 \theta}{2g}}$$

(b) In x direction, it's constant velocity. we know v_{ix} , to find R we need t .

★ Use time to coordinate the motion in two directions.

First step: in y direction, given initial velocity v_{iy} ,
initial and final position = 0, $a = -g$, find t .

Use equation (2) $x_f = \frac{1}{2} a t^2 + v_i t + x_i$

$$0 = -\frac{1}{2} g t^2 + v_{iy} t + 0$$

$$\left(-\frac{1}{2} g t + v_{iy}\right) \cdot t = 0$$

$$t = 0 \text{ or } \frac{2v_{iy}}{g}$$

↑ ↑
start at origin fall back to ground

Second step, use the time in x direction.

$$R = v_{ix} t = v_{ix} \cdot \frac{2v_{iy}}{g}$$

$$\downarrow \quad \begin{cases} v_{ix} = v_i \cos \theta \\ v_{iy} = v_i \sin \theta \end{cases}$$

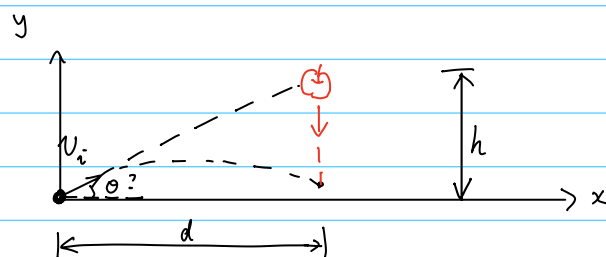
$$\boxed{R = \frac{2 v_i^2 \sin \theta \cos \theta}{g}}$$

* Note that $R = \frac{v_i^2 \sin 2\theta}{g}$, when $2\theta = 180^\circ$ $\sin 2\theta$ is at max = 1,

for a fixed v_i , by tilting the initial velocity angle θ ,

R can reach maximum $\frac{v_i^2}{g}$ when $\theta = 45^\circ$

e.g. 2. A slingshot at ground level is at distance d away from an apple tree.
 slingshot shoots out a ball with initial speed v_i .
 at the same time the apple starts to fall from h above ground.
 At what angle should I aim? (use g as gravitational acceleration)



From $t=0$ to $t =$ the moment the ball hits the apple.

For apple: $y = -\frac{1}{2}gt^2 + h$ (1)

For ball: y direction: $y = -\frac{1}{2}gt^2 + v_i \sin \theta \cdot t$ (2)

x direction: $d = v_i \cos \theta \cdot t$ (3)

3 equations, 3 unknown quantities. Let's get rid of y and t to find θ .
 — same, let's cancel them.

from (1): $-\frac{1}{2}gt^2 = y - h$, put in (2):

$$0 = y - h + v_i \sin \theta \cdot t$$

$$h = v_i \sin \theta \cdot t \quad (4)$$

from (3): $t = \frac{d}{v_i \cos \theta}$, put in (4):

$$h = \cancel{v_i \sin \theta} \cdot \frac{d}{\cancel{v_i \cos \theta}}$$

$$\tan \theta = \frac{h}{d}$$

$$\boxed{\theta = \tan^{-1} \frac{h}{d}} \quad (\text{not depend on } v_i !)$$